convey information via a host
↳ electrochemical pathways

Neural Networks.

↳ also known as ANN

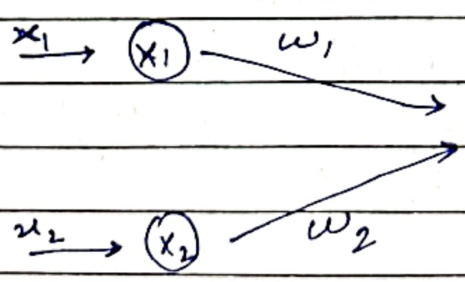
↳ Artificial Neural Network

↳ Parallel Architecture

↳ A computational model inspired by the structure and functioning of the human brain., that learns patterns from data through interconnected artificial neurons (nodes) to solve problems. like pattern matching, classification, optimization, data clustering.

Human Brain	ANN
Neuron	Artificial Neuron (Node)
Dendrites	Input Signals
Synapse	Weight
Cell Body (soma)	Processing Unit
Axon	output Signal

Biological Neurons	Artificial Neurons
Synapses Connect neurons	weights connect nodes
Learns through experience	Learns from training Data
Procers sensory Information	Process Digital data
Makes intelligent decisions	Makes Predictions or classifications.



w_1 and w_2 are Inter-connection Links

Architecture of a simple artificial NN.

$$y_{in} = x_1 w_1 + x_2 w_2 \quad \text{(net input)}$$

$$y = f(y_{in})$$

output = function (net input calculated)

Neuron — or nerve cell, is the basic building block of the nervous system and brain.

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- ANNs posses large no of highly interconnected processing elements called nodes or units or neurons which usually operates in parallel
- Each neuron is connected with the other by a connection link. (forms thousands of links with other neurons)
- Each connection link is associated with weights which contain information about the input signal.

- It should be noted, that each neuron has an internal state of its own. This internal state is called activation or activity level of neuron, which is the function of the inputs the neuron receivis.
- A neuron can send only one signal at a time, which can be transmitted to several other neurons.

Here, X_1 and X_2 are input neurons, which transmit signals
Y is the output neuron, which receive signals.

- Input neurons X_1 and X_2 are connected to the output neuron Y, over a weighted interconnection link (w_1 and w_2)

net input calculated, $y_{in} = x_1 w_1 + x_2 w_2$

where,
x_1 and x_2 are the activations of the input neurons X_1 and X_2

The output y of the output Neuron Y can be obtained by applying activations over the net input
i.e, the function
$$y = f(y_{in})$$

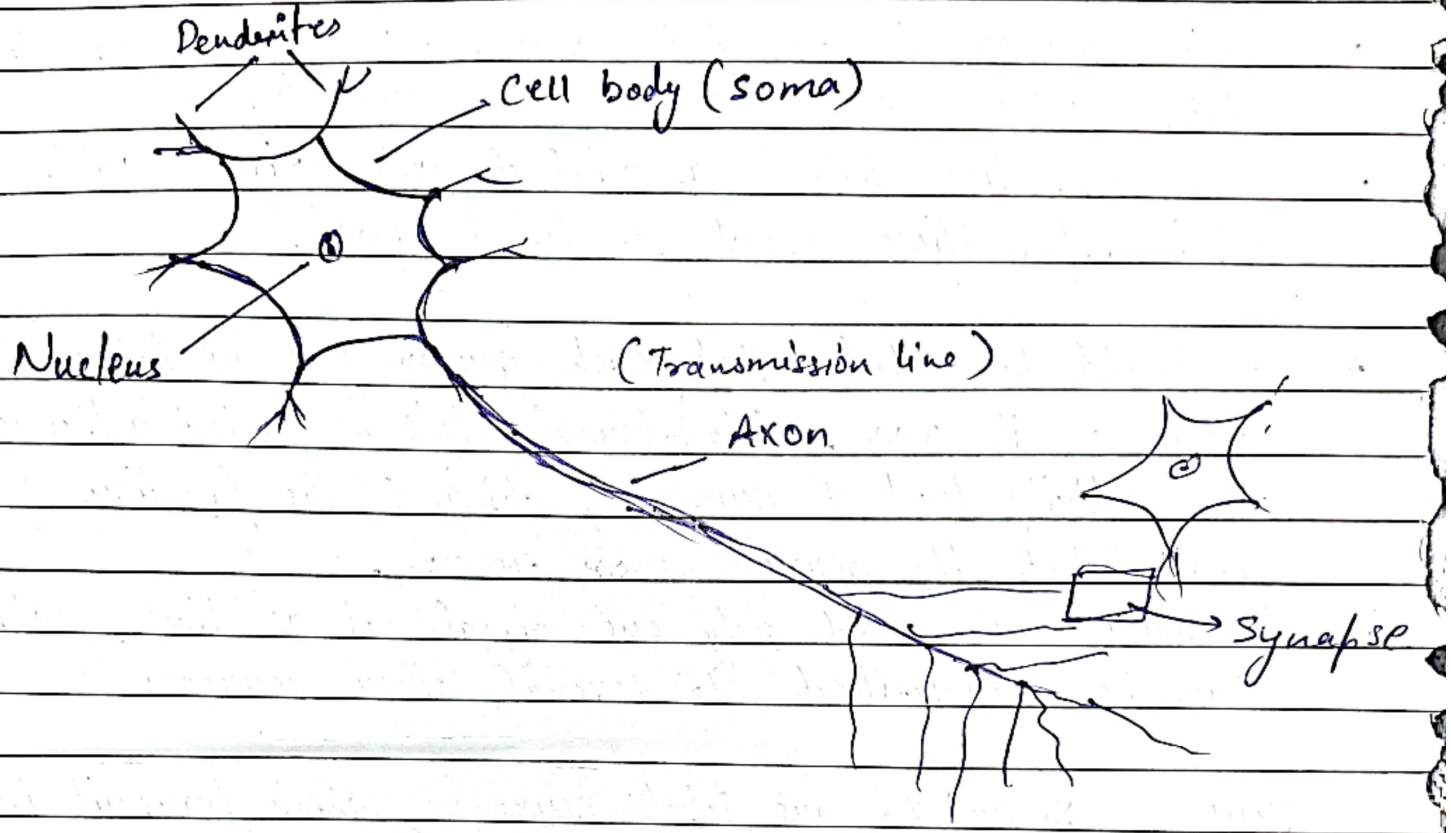
- The function to be applied over the net input is called activation function.

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Biological Neural Network

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↳ Human Brain consists a huge number of neurons approximately 10^{11}, with numerous interconnections.



⊙ There are approximately 10^4 synapses per neuron in the human brain.

Dendrites :- Input
Axon :- output
Soma :- Processing of input / sum of all input signals
Synaptic link :- The electrochemical contact btw neurons.

⊙ It is through synapse that the neuron introduces / transmit its signals to other nearby neurons.

An artificial neuron is a highly simplified mathematical abstraction of a biological neuron

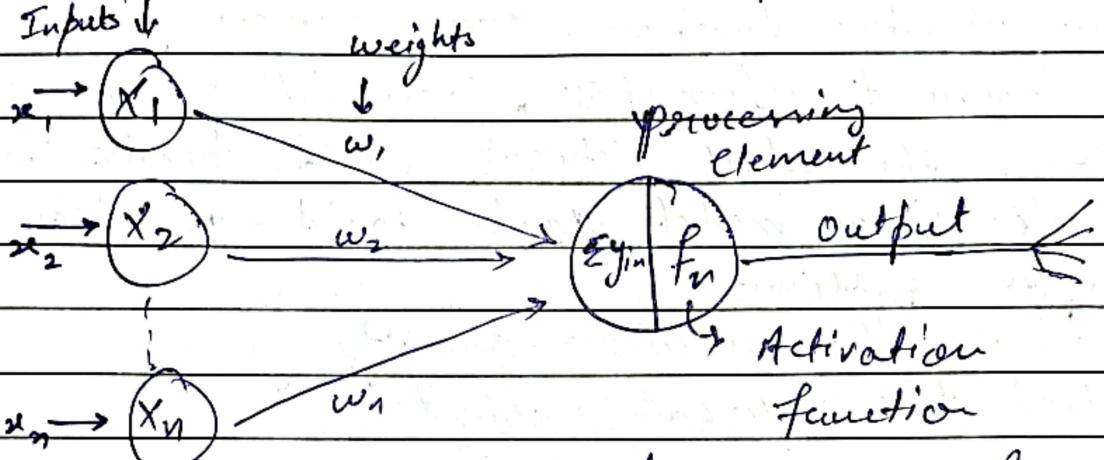
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(signals)

⑥ Electric impulses are passed btw the synapse and the dendrites

Artificial Neural Network / Neuron

↳ Mathematical model of artificial neuron.



Weights :- determine how strongly one neuron influences another neuron.

Biological Neuron	Artificial Neuron
Cell	Neuron
Dendrites	weights / interconnections
Soma	Net input / processing unit
Axon	output

Net input :-

$$y_{in} = x_1 w_1 + x_2 w_2 + \cdots + x_n w_n = \sum_{i=1}^{n} x_i w_i$$

⑧ A NN may be defined as an information processing model that is inspired by the way biological nervous system, such as brain, process information.

Bias :- additional value added to weighted sum of inputs before applying activation function.

↳ control the adjustment / shift of the neuron output

Spiral and improves the flexibility of neural network

Activation Functions

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↳ a mathematical f^n applied to the weighted sum of inputs.

⊙ It determines whether and to what extent a neuron passes its signal to the next layer.

Why activation f^a are needed ?

Without an activation function, every neuron performs only a linear transformation, even with many hidden layers, the entire network would behave like a single linear model and could not learn complex patterns.

⊙ improve the network ability.
⊙ Control the neuron's output

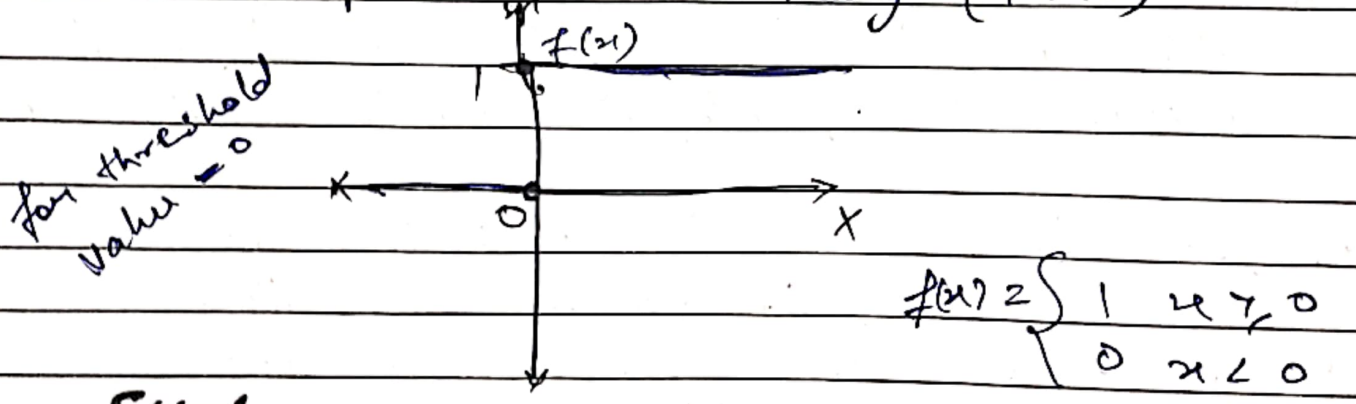
(1) Threshold function :—

↳ also called Binary step function / Hard limit function

$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ 0 & \text{if } x < \theta \end{cases}$$

where, θ = threshold value → minimum input required to fire a neuron

⊙ This f^a is most widely used in single layer network to convert the net input to an output that is binary (1 or 0)



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ex - $x_1 = 1$; $x_2 = 1$ | $w_1 = 0.6$; $w_2 = 0.7$ | $b = 0$ | $\theta = 1$ Suppose

$z = (1)(0.6) + (1)(0.7) = 1.3$

$1.3 \geq 1$ output = $y = 1$

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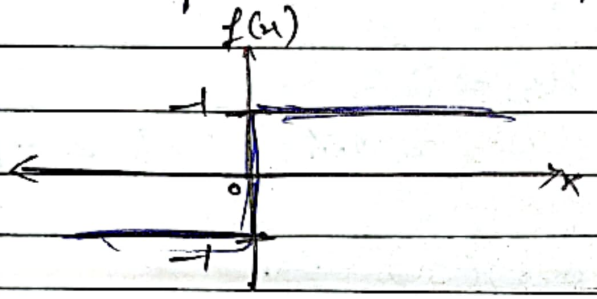
(2) Signum function — Bipolar step function
↳ also known as Sign activation function.

means sign

$$f(x) = \begin{cases} +1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

where, θ = threshold value. $y = sgn(z)$

⊙ This f^n is also used in single-layer nets to convert the net input to an output that is bipolar (+1 or -1)



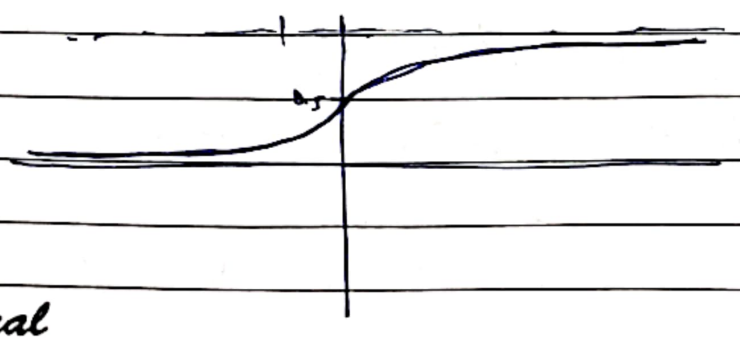
ex $x_1 = 2, x_2 = 3$
$w_1 = 0.5, w_2 = -0.2$
$b = 0.1$
$z = 2(0.5) + 3(-0.2) + 0.1$
$z = 0.5 > 0$
$\therefore y = +1$

NOTE Threshold and signum f^n are mainly used in early neural network models such as perceptrons.

(3) Sigmoid function :- "S" like shape / logistic f^n

It converts any real-valued input into a value between 0 and 1, making it useful for binary classification problems.
limited range

$$f(x) = \frac{1}{1+e^{-x}} = \sigma(x) \qquad 0 < \sigma(x) < 1$$



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when x is very $-ve$. :-

e^{-x} become very large

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$\therefore \quad \sigma(x) \approx 0$.

when $x = 0$:-

$$\sigma(0) = \frac{1}{1+e^{0}}$$

e^{-x} = exponential functions

$$e^{0} = 1$$

$$\therefore \sigma(0) = 0.5$$

when x is very positive :-

$$e^{-x} \approx 0$$

$$\therefore \sigma(x) \approx 1$$

example

$x_1 = 2, \quad x_2 = 3$

$w_1 = 0.4, \quad w_2 = 0.6 \quad b = -1$

$$z = (0.4)(2) + (0.6)(3) - 1$$

$$z = 0.8 + 1.8 - 1$$

$$z = 1.6.$$

$$\therefore y = \frac{1}{1+e^{-1.6}} \approx 0.832.$$

(4) Tan-hyperbolic function :- Tanh (S-shaped)

↳ Hyperbolic Tangent activation function

$$\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

written in terms of sigmoid =

$$\tanh(x) = 2\sigma(2x) - 1$$

output = $\quad -1 < \tanh(x) < 1$

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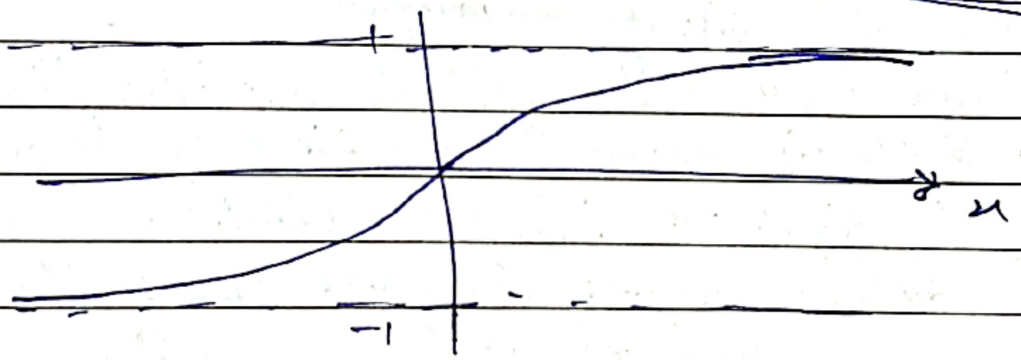
Advantages of NN

(*) NN with the remarkable ability to drive meaning from complicated or imprecise data, could be used to extract patterns and detect trends that are too complex to be noticed by either humans or other computer techniques.

(1) **Adaptive learning** – ability to learn on the base of data given for training or initial experience.

(2) **Self-organization** – creates its own information it recieves during learning time.

(3) **Real time operations** – computations may be carried out in parallel.



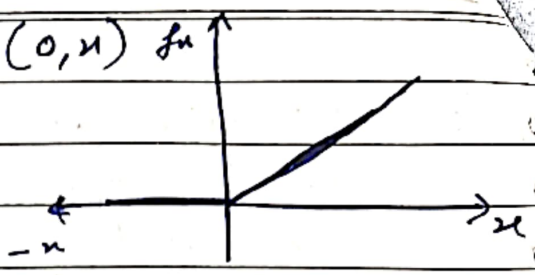
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(5) Ramp function :-

↳ Basic ramp $f^n \Rightarrow f^n = \max(0, x)$ fn

or equivalently $f(x) = \begin{cases} 0 & , x < 0 \\ x & , x \geq 0 \end{cases}$



example,

$x_1 = 3, x_2 = 4, w_1 = 2, w_2 = -1, b = 1$

$z = (2)(3) + (-1)(4) + 1$

$= 6 - 4 + 1 = 3$

Applying the ramp fn, $y = \max(0, 3)$

$y = 3$

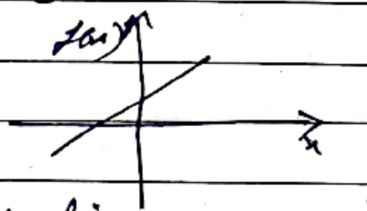
ReLU f^n → Rectified linear unit.

(6) Linear function :- function in which the output changes at a constant rate with respect to the input. → $[-\infty < f(x) < +\infty]$

General form :- $f(x) = ax + b$ → bias

slope

output ↳ input



① called linear because its graph is straight line

↳ word identity means "remaining unchanged".

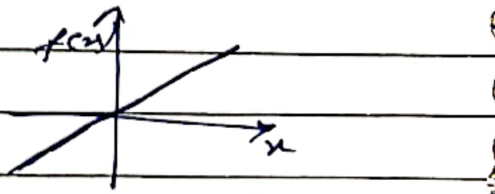
(7) Identity function :- * Simplest activation function

* It returns the same value that it recieves as input.

$f(x) = x$

↳ slope = 1.

Range = $(-\infty, +\infty)$



* Identity function is a special case of linear function.

(8) stochastic function :- → Probability based activation fn with randomness

output of a neuron is probabilistic (random) rather than completely deterministic

* In this function, the same input can produce different outputs according to a probability distribution.

Spiral

For a neuron, first calculate the weighted sum –

$$z = \sum_{i=1}^{n} x_i w_i + b$$

→ In the deterministic activation function –

$$y = f(z)$$

Here, the output y is fixed for a particular z.

→ In stochastic activation function –

$$y \sim P(y|z)$$

Here, the output is sampled according to a probability determined by z.

ANN Architecture :– describes how neurons are arranged and how information flows thru them

- Based of Layer.
  - Single Layer
  - Multi layer.
- Based on Information flow
  - Feed Forward.
  - Feedback / Recurrent.
    - Fully Recurrent.

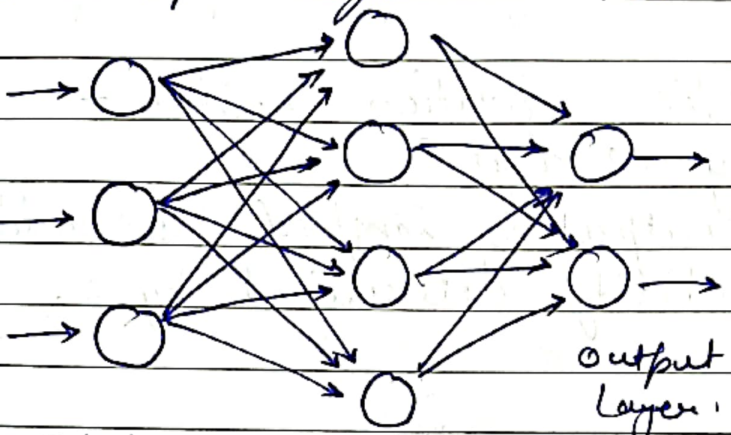
Basic ANN Architectures Models

① Feed Forward Network. e.g.

↳ Information flows in a single direction i.e, from the input layer through hidden layers to the output layer "without loops or feedback"

Applications

* Classification
* Pattern Recognition
* speech Recognition



↳ responsible for learning the complex pattern in the data.
↳ one or more hidden layer.

↳ used for pattern recognition like image and speech classification.

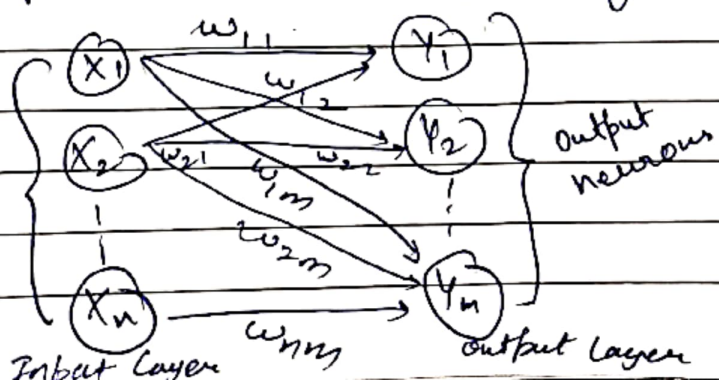
A neural network can have :-

- Zero hidden layers :- Single layer feed forward Netw
- one hidden layer :- shallow neural network
- Multiple hidden layer :- deep neural network.

Types of Feed Forward Network.

① **Single Layer FFN :-** contains an input layer and output layer but no hidden layer

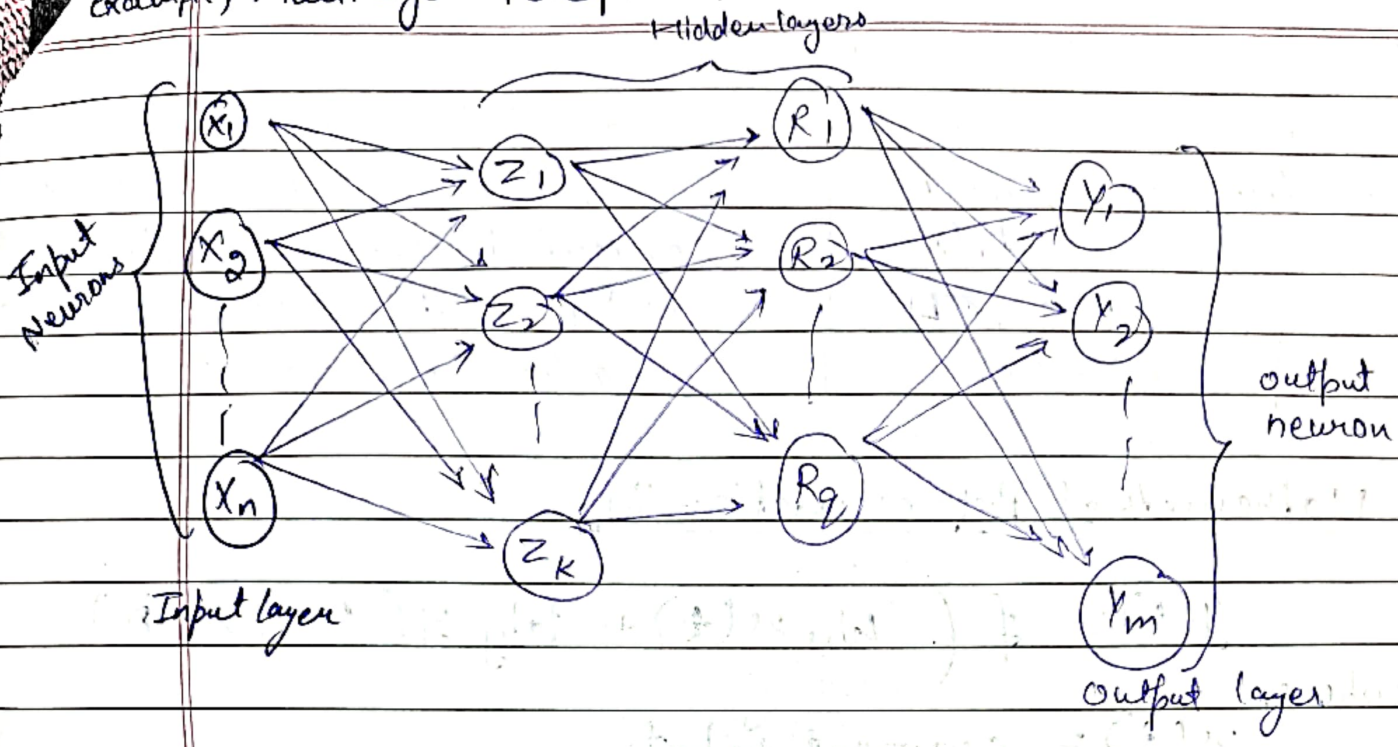
example- Perceptron



- can only solve simply, linear seperation problems.

② Multi Layer FFN :- Contains one or more hidden layers.

example, multi layer Perceptron.



⊙ The hidden layer is internal to the network and has no direct contact with the external environment

⊙ More the no of hidden layers, more is the complexity of the network

↳ also called "Recurrent Neural network architecture"

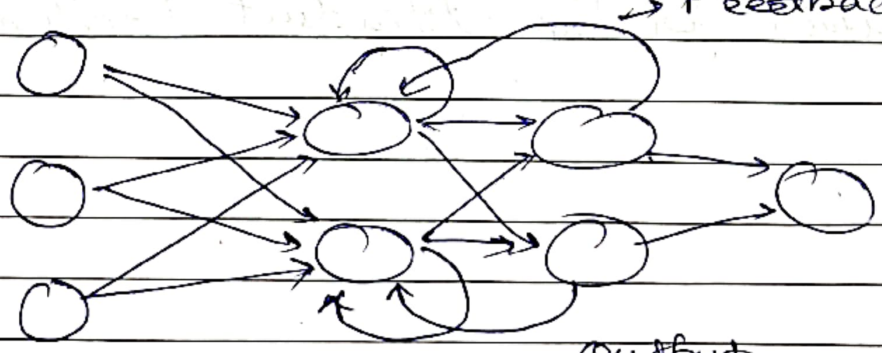
③ Feed Back Network :- network in which the output of a neuron on layer can be fed back to the same layer on a previous layer.

RNN

⊙ contains loops or cycles on feedbacks.

Applications

* Time series Prediction
* Signal Processing



⊙ Suitable for sequential data.

Features	Feed Forward	Feed Backward
Direction	Only Forward	Includes feedback connections
Loops / cycles	No	Yes
Memory	No state	Has state / memory
Previous information	No influence	Can influence future output
Suitable for	static data	sequential / dynamic data
Example	Perceptron, MLP	RNN, Hopfield Networks, LSTMs
Architecture	Acyclic	Cyclic / Recurrent

Mathematical Representation of

$$y(t) = f\left(W_x x(t) + W_y y(t-1) + b \right)$$

where,

- $x(t)$ = current input
- $y(t-1)$ = previous output
- W_x = input weight.
- W_y = feedback weight
- b = bias.
- f = activation function
- $y(t)$ = current output.

Feedback $\neq$ back propagation

The feedback weight determines how strongly the previous state/output influences the current calculations.

(D) suitable for time dependent problems.